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Students' struggle with digital addiction: the truth of brain rot

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Abstract

Background Due to the high level of digital addiction, the term *brain rot* was chosen as the word of the year for 2024 determined by a public vote conducted by Oxford University, and it has been used by experts to express the effect of consuming an excessive amount of low-quality online content, especially on social media. This study investigates the prevalence of brain rot among higher education students in a faculty of education, explores its underlying causes, examines its implications for education, and identifies potential preventive measures.

Methods A qualitative case study design was employed to gain an in-depth understanding of the phenomenon. The study group comprised 113 students enrolled in various departments of the Faculty of Education at a public university in Türkiye during the 2024 to 2025 academic year. Participants were selected using criterion sampling, a purposive method designed to identify information-rich cases. Data were collected through observations, semi-structured interviews, and document analysis, and analyzed using content analysis.

Results The findings indicate that nearly all participants experienced symptoms of brain rot, which were attributed to emotional factors, habitual behaviors, and environmental, social, and cognitive influences. It has been determined that brain rot primarily causes students to experience attention deficits, learning difficulties, mental fog, diminished cognitive performance, reduced academic achievement, weakened social relationships, reluctance to attend school or engage in educational activities, and decreased time efficiency. Participants recommended several strategies to mitigate brain rot, including educational interventions, environmental modifications, digital usage regulation, digital detox, and personal self-regulation practices.

Conclusions The results highlight how students' experiences of brain rot are reflected in their educational lives and underscore the measures that need to be taken to prevent its occurrence.

Keywords Digital addiction, Internet addiction, Social media addiction, Brain rot

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Introduction

The widespread integration of computers, the internet, smartphones, and digital platforms into daily life has transformed how individuals engage in communication, learning, and leisure. These tools are extensively used across all age groups for diverse purposes [1]. However, their pervasive use, particularly among students, has also given rise to various risks, most notably digital addiction [2]. Defined as a dependence on multiple forms of digital media, including the internet, digital games, smartphones, and social networking platforms [3], digital addiction is widely recognized as a modern behavioral disorder [4].

Users are drawn to digital media for numerous reasons, including identity construction, entertainment, escapism, information seeking, self-expression, interpersonal interaction, stress relief, and staying current with trends [5–10]. Moreover, platform architecture, such as infinite scrolling, content flow algorithms, and gamified features, has been shown to reinforce compulsive use [5, 11].

Research increasingly links digital addiction to a range of psychosocial and academic problems, including stress, social anxiety, depression, loneliness [12, 13], insomnia, eating disorders [14], reduced academic motivation [15] and declining academic performance [16]. Studies also highlight its role in attention deficits, impaired social relationships, and physical symptoms such as back pain and fatigue [2, 17, 18].

While digital addiction has been extensively studied [19–22], a newly emerging and less explored phenomenon, brain rot demands urgent academic attention [23]. Voted the Word of the Year in 2024 by Oxford University Press, brain rot is broadly defined as the cognitive, emotional, and behavioral deterioration resulting from overexposure to low-quality online content, particularly via social media [24]. Though indirectly foreshadowed by Henry David Thoreau's [25], reflections on mental stagnation in *Walden*, the term has recently gained traction to describe the cognitive fatigue and superficiality characterizing digital life in the 21st century [23].

Brain rot is associated with broader constructs such as digital, internet, and social media addiction [26–28]. Recent scholarship points to its adverse impact not only on individuals' cognitive and emotional development but also on educational and societal systems at large [29, 30]. As global concern about its effects grows, scholars have emphasized the need for structural educational reform to counteract its influence [23]. Brain rot should not be understood as a type of addiction. Rather, it refers to a condition that arises as a consequence of digital, internet, and social media addictions. Conceptually, while addiction denotes a behavioral dependence characterized by compulsive engagement and withdrawal symptoms, brain rot represents the cognitive and affective deterioration

that follows such addictive engagement. Grounded in neuroscientific evidence, brain rot manifests as mental exhaustion and a measurable decline in cognitive functions and attentional capacity resulting from prolonged exposure to excessive digital content [31–33]. At the same time, brain rot is an important problem that also affects the educational process [16]. Therefore, solutions need to be found on a global scale [23]. Teachers are faced with the task of combating the threat of brain rot.

In addition, pre-service teachers play a pivotal role in shaping society's digital awareness and literacy. As future educators, their attitudes and behaviours towards digital technologies will significantly influence how upcoming generations engage with digital media. Therefore, examining the digital habits and awareness levels of this group is crucial, not only for understanding their personal development but also for ensuring a more digitally responsible and conscious educational culture. As future educators, their awareness of brain rot, both in theory and in practice, is critical to addressing its educational implications. For this reason, this study focuses on pre-service teachers to explore how brain rot affects their learning experiences and academic functioning. The research also seeks to identify contributing factors and possible interventions.

Specifically, the study aims to (1) examine pre-service teachers' awareness of brain rot and digital addiction, (2) explore the perceived effects of brain rot on educational engagement, readiness, and productivity, and (3) identify their suggestions for preventing and mitigating the phenomenon. By addressing a timely and underexplored topic, this study contributes to the emerging literature on brain rot and offers insights for educational stakeholders seeking to promote healthier digital habits.

Method

This section outlines the research model, study group, data collection tools, data analysis, and validity-reliability measures.

Research model

The present study was conducted using a case study design. Case studies are research designs that aim to collect in-depth information about real-life, contemporary situations and to present the examined case through identified themes [34]. This design is frequently preferred in practice-oriented fields such as education [35]. In this context, the case examined in the current research is brain rot. Since the study aimed to reveal how students experience brain rot, the reasons behind it, and how this situation reflects on their educational processes, a case study design was deemed suitable. Case studies aim to obtain detailed information about the investigated situation by utilizing various data sources such as

observations, interviews, and documents [35]. In this research, these three techniques were used to capture the depth of the situation. To ensure the data were not limited or biased and to achieve data triangulation, multiple data collection techniques were employed [34].

Patton [36] emphasizes that, to conduct an in-depth analysis of a case, it is essential to gather detailed perspectives from different individuals related to the situation. Accordingly, in this study, the observations and opinions of university students from different departments who experience brain rot were examined in detail.

Study group

The study group comprised 113 pre-service teachers enrolled in the Faculty of Education at Kahramanmaraş Sütçü İmam University, a public university in Türkiye, during the 2024 to 2025 academic year. Criterion sampling, a purposive method targeting information-rich cases aligned with research objectives [36], was used to select students who (1) were enrolled in teacher education programs and (2) reported experiencing symptoms of brain rot [36]. Ethical standards were strictly followed. Participants' identities were anonymized using codes (K1–K15 for interviews; D1–D113 for documents). As shown in Tables 1 and 72 participants were female and 41 were male, spanning six departments: early childhood, elementary, science, English language, social studies, and Turkish language education.

Students reported an average of 6 h and 30 min of daily unproductive screen use, with only 32 min and 30 s of productive use. Face-to-face interviews were conducted with 30 students selected from a total of 113 participants. In qualitative research, the sample size is determined by the point at which data saturation is reached [37]. Accordingly, the number of interviews in this study was established based on the principle of data saturation. Data saturation was considered to have been achieved when no new codes or themes emerged during successive interviews and participant responses began to repeat. At this point, the sample was limited to 30 participants. Moreover, it was observed that this number of interviews

ensured sufficient depth, variation, and richness of data to address the research objectives comprehensively.

Table 2 shows that the interviews were conducted with 30 university students studying in 6 different departments of the Faculty of Education. These participants are students in the departments of preschool, classroom, science, English, social studies, and Turkish teaching. 20 of the participants are female, and 10 are male. Their ages are concentrated in the 19, 20, and 21 age groups. According to Table 2, almost all participants mostly use screens inefficiently, with inefficient screen time averaging 4 h and 38 min per day. The participants' average daily use of screens efficiently is 1 h and 36 min. It is also observed that K3, K10, K19, and K25 use screens more efficiently, while K29 uses screens equally efficiently and inefficiently. During the interviews, students were asked again about their efficient and inefficient screen use. Many reported spending considerably more time on screens for non-educational or unproductive purposes than indicated in the documented screen-time records. It was determined that the screen time reported in the participant observation form differs from the screen time presented in the interviews. This striking difference may be related to the students being able to confront themselves more openly while writing and not specifying names in the document.

Data collection instruments

Three tools were used to gather research data in the study: observation forms, participant documents, and semi-structured interviews.

Observation

Researcher and participant observation forms were developed by the researcher. The former tracked digital usage over three months by instructors; the latter documented six days of individual screen use. Of 140 distributed forms, 113 valid responses were analyzed.

Table 1 Demographic characteristics and reported Screen-Time of participants

Department	Participants	Female	Male	Average Unproductive Screen Time	Average Productive Screen Time/Day
Preschool Ed.	22	17	5	7 h	45 min
Elementary Ed.	21	14	7	6 h	30 min
Science Ed.	5	4	1	6 h	30 min
English Language Ed.	25	13	12	7 h	30 min
Social Studies Ed.	20	11	9	7 h	45 min
Turkish Language Ed.	20	13	7	6 h	15 min
Total	113	72	41		
TOTAL SCREEN TIME				6 h 30 min	32 min 30 s

Table 2 Information on interviewed participants

Participant	Department	Class	Age	Gender	Daily Unproductive Screen Time	Daily Productive Screen Time
K1	Preschool Education	1	19	K	4 h	1 h
K2	Preschool Education	1	20	K	5 h	1 h
K3	Preschool Education	1	20	K	1 h	3 h
K4	Preschool Education	1	20	K	6 h	1 h
K5	Preschool Education	2	20	K	6 h	1 h
K6	Preschool Education	2	21	K	7 h	1 h
K7	Preschool Education	2	21	E	6 h	1 h
K8	Preschool Education	2	21	E	6 h	1 h
K9	Preschool Education	2	21	E	6 h	1 h
K10	Preschool Education	1	35	E	1 h	3 h
K11	Elementary Education	2	20	K	4 h	1 h
K12	Elementary Education	2	20	K	4 h	1 h
K13	Elementary Education	2	21	K	6 h	2 h
K14	Elementary Education	2	21	K	6 h	1 h
K15	Elementary Education	2	20	K	5 h	2 h
K16	Science Studies Ed.	1	19	K	5 h	2 h
K17	English Language Ed.	4	24	K	5 h	1 h
K18	English Language Ed.	2	21	K	6 h	30 min
K19	English Language Ed.	2	19	K	30 min	3 h
K20	English Language Ed.	2	20	E	5 h	1 h
K21	English Language Ed.	2	22	E	7 h	1 h
K22	Social Studies Ed.	1	19	K	5 h	2 h
K23	Social Studies Ed.	1	20	K	6 h	-
K24	Social Studies Ed.	1	19	E	6 h	2 h
K25	Social Studies Ed.	1	19	E	2 h	6 h
K26	Turkish Language Ed.	1	20	K	4 h	2 h
K27	Turkish Language Ed.	1	18	K	4.5 h	30 min
K28	Turkish Language Ed.	1	19	K	4 h	1 h
K29	Turkish Language Ed.	1	18	E	3 h	3 h
K30	Turkish Language Ed.	1	18	E	6 h	2 h
TOTAL SCREEN TIME					4 h 38 min	1 h 36 min

Documents

Documents are written materials providing information about phenomena, behaviors, or situations [37]. Participants were asked to write reflective narratives on their brain rot experiences. To ensure relevance, the document prompts were aligned with research sub-questions and reviewed for bias and completeness [37, 38]. The narratives explored how students conceptualized brain rot, described their experiences, identified its causes and educational impacts, and proposed potential solutions.

Interviews

Semi-structured interview protocols were developed through literature review and expert input. A pilot with five students refined the questions. Of the 113 participants, 30 consented to interviews. All instruments were carefully archived after data collection.

Data collection procedure

As outlined in Fig. 1, the study began with a literature review, followed by selection of 140 students.

Observation forms were administered in six classes, and expert feedback was used to refine instruments. Interviews were conducted with 30 students, and 113 documents were analyzed. Based on these inputs, a participant observation form was created and validated by experts.

Data analysis

The analysis phase is the stage in which the data are interpreted, comparisons are made, and a framework for comments is created [36]. In the analysis of the data of this study, the content analysis technique and the inductive analysis method were adopted [36]. Data were analyzed in three phases. First, the data obtained through the participant observation form were tabulated. The table included each student's daily screen time and the efficiency status of this time. The researcher observation form data were used to diversify and substantiate the data obtained through the document form and the interview form within the framework of the research findings. In the second phase, documents from 113 students were analyzed using content analysis. In the third phase,

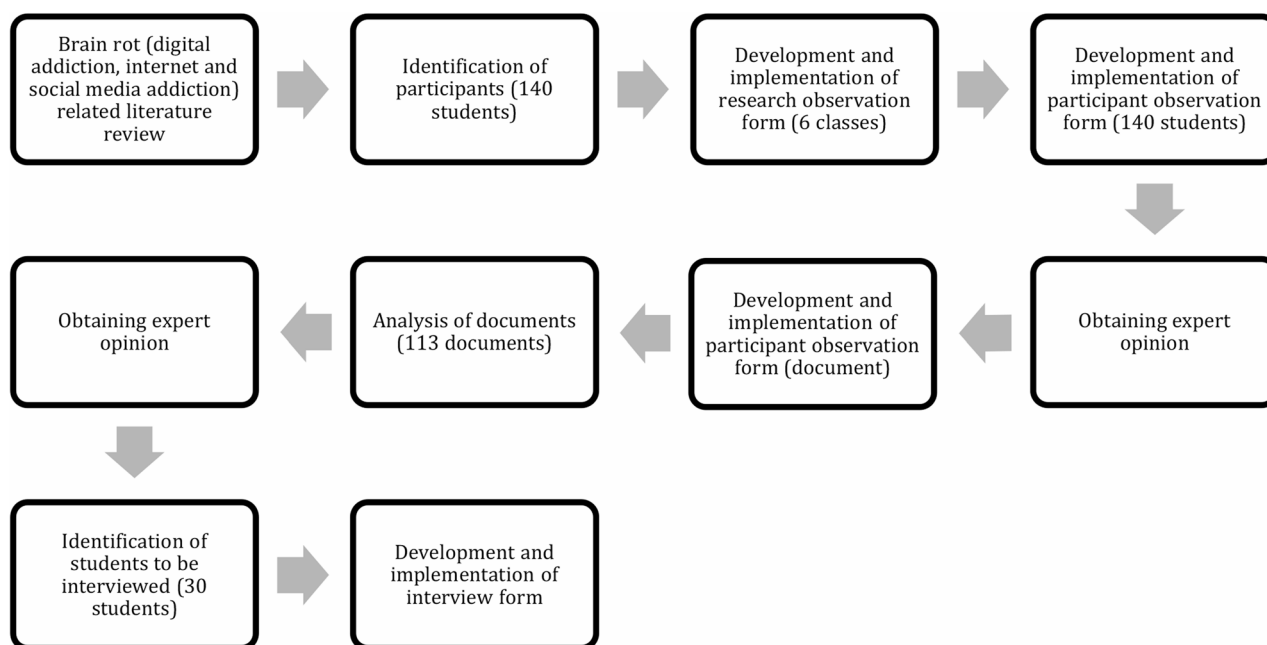


Fig. 1 Data collection process

the interview data of 30 students were analyzed. In the content analysis process, codes were first extracted, and then categories and themes were created based on these codes [34–36]. In this study, the coding was carried out separately by the researcher and a faculty member specializing in educational sciences at the same university. The agreement between the coders was calculated using Miles and Huberman's formula ($\text{Reliability} = \frac{\text{Agreement}}{\text{Agreement} + \text{Disagreement}} \times 100$) [39]. The agreement between the researcher's and the expert's codes was found to be 0.91. In cases where there were different codes, the codes were reviewed again, and the researcher and the expert discussed and decided on the new code [40]. In the final phase, interview transcripts from 30 students were similarly coded. Discrepancies between coders were resolved through discussion [40]. Findings were supported by direct quotations from participants, and all data sources were integrated to construct a comprehensive understanding of the phenomenon.

Validity and reliability

In qualitative research, ensuring validity and reliability is predicated on conducting the study ethically [35]. To carry out this research ethically, the recommendations of Christensen, Johnson, & Turner were followed, giving careful attention to participants' voluntary involvement and providing students with detailed information about the study [41]. This research was approved by the Kahramanmaraş Sütçü İmam University Social and Human Sciences Ethics Committee (Decision No. 2025-21, dated 21.03.2025). Participants included in the study

were provided with detailed information about the purpose, process, confidentiality principles, and voluntariness of the study. Written informed consent was obtained from all participants, and they were informed of their right to withdraw from the study at any time. Participants' identities were kept confidential, with codes such as Ö1, Ö2 for participants and D1, D2 for documents [42]. Additionally, efforts were made to faithfully reflect the research process. To ensure the validity and reliability of the study, the four strategies recommended by Lincoln & Guba for qualitative research credibility, transferability, dependability, and confirmability were employed [43]. The work conducted within this framework is summarized below.

Credibility refers to the accuracy of findings and was ensured through triangulation, prolonged engagement, participant validation, expert review, and researcher reflexivity [34, 36]. Data triangulation in this study was achieved using observations, interviews, and document analysis, enhancing the depth and consistency of findings.

The researcher engaged with participants from September 2024 to March 2025, observing them in natural settings and collecting in-depth data. Two experts reviewed the instruments, and participant validation was obtained by confirming the accuracy of data with participants. The researcher's role was shared transparently with participants to minimize bias and increase trust.

Generalizability is not a goal in qualitative research; instead, transferability refers to the extent to which findings can be applied to other contexts [35]. To support

Table 3 Six-day digital tool usage data of a sample participant

Day	Unproductive screen time	Productive screen time
Monday	8 h (Social media, shopping websites)	15 min (Academic studies)
Tuesday	6 h (Social media, gaming)	30 min (Research)
Wednesday	5 h (Social media, gaming)	20 min (Homework)
Thursday	6 h (Social media, gaming)	15 min (Homework)
Friday	5 h (Social media, shopping websites)	-
Saturday	7 h (Social media, gaming)	30 min (Research)

transferability, the research process from sampling to data collection and analysis was described in detail, following the recommendations of Lincoln and Guba [43]. Findings were presented with direct participant quotations, and language was kept clear and accessible. Additionally, tables and figures were used to help readers evaluate the applicability of results in their own contexts.

Dependability concerns the consistency of findings. To ensure this, the researcher and an expert independently coded the data and developed themes. Discrepancies were resolved through discussion, strengthening the reliability of results [34, 40].

Confirmability addresses the objectivity of the research and was achieved by grounding all interpretations in participant data. An audit trail of the entire research process was maintained, ensuring transparency and enabling external verification [34]. Direct quotations were used to reduce researcher bias and ensure that conclusions reflected participants' authentic views. All procedures were also documented systematically to allow future researchers to track the methodological steps taken.

Findings

This section presents the results derived from observations, documents, and interviews, organized around the study's objectives: participants' experiences with brain rot, its causes, educational impacts, and suggested prevention strategies.

Participants' experiences of brain rot

Findings from all data sources indicated that all 113 participants experienced symptoms of brain rot. Table 3 presents one participant's six-day digital usage, showing 37 h of unproductive screen time compared to only 1 h and 50 min of productive use. Reflecting on this, the participant stated:

I didn't know I was this addicted. After observing myself for six days and filling out this form, I realized that I need to take precautions. No wonder I have been unable to focus on my studies and research. (K4)

Table 4 Findings on the perceived causes of brain rot among participants

Theme	Codes	Doc. (f)	Int. (f)
Emotional causes	Boredom/desire for relaxation	30	12
	Avoiding tasks/desire for inaction	28	10
	Feeling lacking when not on social media	10	8
	Escaping from real problems	8	6
	Feeling happy on social media	3	2
Habitual causes	Wanting to spend time	24	8
	Spending fun time	20	6
	Perceived sense of rest	12	5
	Desire for convenience	4	2
	Home confinement during the pandemic	10	15
Environmental causes	Influence of social surroundings	8	3
	The allure of digital content	4	2
	The digitalization of education	3	2
	Slowness of real life	2	3
	Easy access	2	2
Social causes	Peer pressure	2	1
	Desire for individuality	10	4
	Meeting the need for communication	7	7
	Being seen on social media	8	3
	Desire for validation	6	1
Cognitive causes	Mental exhaustion	8	2
	Curiosity	5	2
	Quick access to information	4	3

In a classroom observation on December 22, 2024, when assigned a research task during a philosophy of education course, the student instead accessed a shopping site on their phone. On December 15, she was observed using their phone throughout the class break.

Causes of participants' brain rot

This section presents the findings regarding the underlying causes of participants' experience of brain rot, based on thematic analysis. As shown in Table 4, these causes are grouped under the themes of emotional causes, habitual patterns, environmental causes, social causes, and cognitive causes. Among these, emotional factors were reported as the most prominent contributors.

Participants frequently identified boredom and the desire for relaxation as key emotional drivers. Many noted that they turned to digital media to escape monotony or to find temporary relief. This is illustrated in K14's reflection:

I get bored easily... I use my phone to relax, but I guess that's what's rotting my brain.

Similarly, avoiding tasks or seeking inaction was commonly reported as a coping strategy to escape academic or personal responsibilities (K11, D35). Classroom observations supported these accounts; during a session on

December 9, 2024, students appeared disengaged and were seen using their phones, suggesting task avoidance through digital distraction.

Another emotionally driven cause was the feeling of lack or disconnection when not using social media. Participants expressed concerns about missing out, feeling incomplete or anxious if not constantly online. One participant explained:

When I don't pick up my phone, I feel like I won't be able to keep up with current events. (K29)

Some participants also identified emotional motivations behind brain rot, including a desire to escape real-life problems and a need to feel happy in their following statements:

When I go through negative experiences in life, I use my phone to forget about them. (K5)
Some participants, on the other hand, stated that they act based on the feeling of happiness. This situation is reflected in K29's statements:
When I'm online, I feel happy, like I disappear. (K29).

In a subsequent observation conducted on December 10, 2024, students were again seen spending their breaks on their phones, often smiling as they did so, suggesting a momentary sense of enjoyment or relief during use.

In terms of habitual causes, many participants mentioned that they engaged with digital content simply to pass the time or as a form of entertainment and rest. K20 stated: "I have too much free time, I just want to kill it," while K23 described their use as driven by "a need for entertainment." Others framed it as a perceived sense of rest (D7). The desire for convenience was also evident, as expressed by a participant:

I prefer not to think, I choose convenience, I don't question, I take the easy way out, I've lost my reading habit. (D23)

Among environmental causes, the COVID-19 pandemic was frequently cited as a significant contributor to increased experiences of brain rot. As one participant explained:

The pandemic period was very impactful. We were locked at home during the pandemic, we had nothing to do, so we clung to our phones. (K11)

Participants also highlighted other environmental influences such as peer pressure (D76), the perceived slowness of real life (D18), and the appeal and easy accessibility of

digital content (D18). Finally, the digitalization of education was noted as a contributing factor. A 20-year-old female participant described how academic use of digital tools often led to distraction:

Our teachers give us assignments, so I have to check my phone and research on the internet, but then I end up drifting off to other things. (D12)

This pattern was also evident during an observation on December 17, when students were encouraged to use AI tools for a class activity. While conducting their research, many were observed switching between academic platforms and social media, reflecting a blurred boundary between purposeful engagement and distraction.

In relation to social causes, many participants highlighted a desire for solitude as a contributing element to their experience of brain rot (D23). Others pointed to the need for communication (K6), the urge to remain visible on social media (K13), and the desire for peer approval (K17). For instance, some participants reported going online initially to connect with others, only to find themselves overwhelmed or distracted (K20). Others expressed that they felt invisible unless they were active on social platforms. As one participant explained:

Everyone sees me through my social media. If I don't post something, if I don't log in, it feels like no one will see me. (D26)

Within the scope of cognitive causes, participants identified mental exhaustion, curiosity, and the perceived efficiency of quick information access as key contributors to their experience of brain rot (D109). Several described feeling too mentally fatigued to engage in sustained thinking, leading them to seek cognitive relief through digital media. Although this initially served to satisfy their curiosity or provide a sense of rest, many reported becoming passively absorbed in unrelated or trivial content. As one participant put it:

I feel mentally exhausted, I just want to rest, and this pulls me into it. (K15)

This behavioral pattern was also evident during an observation on October 08, 2024 in a philosophy of education class, where most students, when posed with reflective questions, turned to phone applications rather than engaging in independent thought.

Findings on the effect of the participants' brain rot on their education

This section presents the effects of participants' brain rot experiences on their education. As displayed in Table 5,

Table 5 Findings on the impact of brain rot on participants' educational readiness and productivity

Theme	Category	Codes	D (f)	I(f)
Impact on educational readiness	Emotional impact	Attention deficiency	92	74
		Deterioration of mental health (mental fog)	57	5
		Falling into depression	20	4
		Feeling anxious and stressed	18	2
		Sense of emptiness	13	4
		Burnout	3	2
		Loss of self-confidence	3	1
		Withdrawal/feeling lonely	2	1
		Intolerance/Impatience	2	1
	Cognitive impact	Inability to focus on lessons	102	19
		Causing cognitive decline	57	5
		Experiencing mental fatigue	16	7
		Loss of motivation to learn	14	6
	Physical impact	Drifting away from science/manipulation	2	1
		Experiencing sleep problems	11	5
		Experiencing head and neck problems	2	6
		Experiencing eye health problems	2	6
		Experiencing muscle and skeletal problems	2	4
		Being overweight	4	1
		Experiencing excessive fatigue	3	2
Impact on educational productivity	Impact on cognitive efficiency	Negatively affecting thinking skills	64	12
		Experiencing forgetfulness	20	9
		Losing the habit of reading	16	13
		Not studying or experiencing problems (regarding duration, depth, form, procrastination)	12	16
		Decline in academic success	16	4
		Having difficulty understanding	10	2
		Having difficulty learning	9	2
		Experiencing problems in problem-solving	9	1
		Inability to fulfill responsibilities	8	6
	Impact on social efficiency	Failure to actualize self-potential	8	6
		Weakening of social relationships	9	13
		Experiencing problems in communication with friends	4	3
		Not participating in educational activities	3	1
		Showing reluctance to attend school	2	1
	Impact on time efficiency	Experiencing time loss	24	11
		Losing the perception of time	8	4

the findings are categorized under two main themes: impacts on educational readiness (affective, cognitive, and physical effects) and impacts on educational efficiency (cognitive, social, and time efficiency).

Under the educational readiness theme, the most pronounced effects of brain rot appeared in participants' affective states. Participants consistently described difficulties with attention regulation and mental clarity. For instance, D17 noted, "I'm an unmindful user. I spend so much time on screens that my brain feels like it's stopped. I experience constant attention deficits in class." Similarly, K12 and K27 described feelings of cognitive stagnation and emotional numbness, suggesting a diminished capacity for sustained engagement. These challenges were also apparent during an observation on October 15, 2024, in which students were frequently observed checking their phones during a lesson presentation, indicating fragmented attention and reduced focus.

Emotional impacts of brain rot were prominently reported and manifested in various interrelated forms. A recurrent theme across participants was attention deficiency and cognitive sluggishness, which often accompanied broader mental health deterioration, including what several described as "mental fog." K26 characterized this condition as an addictive state: "I am an addict; I can't do other activities anymore. Sometimes I don't even eat." Emotional exhaustion and burnout were frequently reported, as reflected in D59's admission: "Sometimes I spend so much time on my phone that burnout never leaves me; I can't care for my friends and family or fulfil my responsibilities." Feelings of depression, stress, and anxiety were also common. D35 stated, "My quality of life is quickly decreasing; my anxieties are increasing. I feel sadness and hopelessness." Other participants reported a persistent sense of emptiness (K29), heightened regret and guilt over lost time and academic delays (K2), and a significant decline in self-confidence and self-worth (D46). For example, K8 described feeling disconnected and isolated: "I become disconnected from my surroundings; I feel isolated." Participants also highlighted increased irritability, impatience, and a reduced capacity for emotional regulation. One participant stated in the documents that they have particularly become intolerant. This situation is reflected in D22 as: "I no longer have patience or tolerance for anything."

Cognitive impacts of brain rot were prominently reflected in participants' reports of diminished educational readiness. A recurring issue was the inability to maintain focus during lessons, which participants frequently attributed to mental fatigue and cognitive overload (K13). Several students also described signs of cognitive decline, including reduced intellectual engagement and difficulty processing or retaining new information. One participant explained this issue:

I am experiencing mental exhaustion. I've stopped questioning information, lost interest in different perspectives, and feel my capacity to learn is decreasing. (D17)

Additionally, it has been determined that brain rot causes problems in students such as losing learning motivation and distancing themselves from science, with students reporting a weakening desire to read, study, or engage critically with course material (K24). Another effect of the brain rot involved the internalization of misinformation and reduced critical thinking (D71). Likewise, several participants described drifting away from scientific reasoning and becoming influenced by unverified content encountered online (D23). These findings suggest that brain rot not only undermines cognitive focus but also weakens students' motivation, critical capacity, and alignment with scientific inquiry.

Physical symptoms, particularly those associated with disrupted sleep, were also prevalent. K23 remarked that irregular sleep patterns negatively affected their ability to follow lessons. Several participants also reported headaches, musculoskeletal pain, fatigue, and weight gain, as illustrated by K29 and D45. These observations were corroborated by classroom monitoring on 22 October 2025, during which students remained physically inactive, using their phones throughout an extended class break.

In terms of educational productivity, participants emphasized the adverse impact of brain rot on cognitive functioning, memory retention, and academic performance. K11 linked the decline in higher-order thinking skills to digital overexposure, while D53 described difficulties with information recall as detailed in his following remarks:

After reading 5–10 pages of a book, my hand reaches for the phone, and when I pick the book up again, I have forgotten what I read. It happens in class too, as soon as I leave class, I look at my phone, hours pass, then I wonder what we studied in class today, and there's nothing....

Many participants also emphasized that brain rot had disrupted their reading habits and studying routines, citing reduced duration, depth, and consistency of effort, as well as chronic procrastination. The cumulative effect of these issues was a noticeable decline in academic achievement. Students reported difficulties in understanding course material, absorbing new information, and applying their knowledge (K3, D5). Others cited problems in problem-solving tasks, an inability to manage academic responsibilities, a sense of underachievement and unrealized potential (D27, D18). This pattern was evident during a class activity on 5 November 2024, where

students struggled to recall previously covered content and immediately turned to their phones for answers. These accounts indicate that brain rot not only compromises students' cognitive efficiency but also obstructs their academic growth and personal development.

The findings also indicate that brain rot significantly undermines students' social efficiency within educational settings. Many participants reported a weakening of social relationships and a noticeable decline in the quality of their interactions. For instance, D27 likened the experience to "being locked in a dark room," suggesting a sense of social isolation that negatively affected their connections with others. Challenges in communication with peers were frequently noted, particularly during class breaks when most students preferred to remain absorbed in their phones rather than engage in conversation. An observation conducted on November 12, 2024, confirmed this pattern: while a few students interacted socially, the majority isolated themselves with their devices. Furthermore, participants expressed a marked reluctance to attend school and participate in educational activities, often attributing this avoidance to cognitive overload and emotional fatigue associated with excessive digital exposure. This was reflected in a participants' following statements:

My desire to go to school is decreasing. I don't want to do anything related to school that would tire my mind. (D4)

These findings collectively suggest that brain rot not only impairs individual academic functioning but also disrupts students' social engagement and willingness to be actively involved in their educational environments.

Findings on participants' suggestions for preventing brain rot

This section presents participants' suggestions for preventing brain rot, as categorized in Table 6 under four main themes: providing education, organizing the environment, usage control, and the developing individual alternatives.

Across all themes, participants consistently emphasized the critical role of education. They advocated for targeted training on digital addiction, digital literacy, the harms of brain rot, and conscious technology use. As one participant stated, "There should be awareness and sensitization training on addiction and brain rot" (K13), while another highlighted the importance of media-based outreach: "Information and awareness training should be given in the media on how to use technology appropriately without becoming addicted" (D13).

Another prominent theme involved organizing educational and social environments to discourage excessive

Table 6 Findings on participants' suggestions for preventing brain rot

Theme	Codes	Doc. (f)	Int. (f)
Providing Education	Digital addiction awareness training should be provided	29	12
	Education on the harms of brain rot should be given	29	12
	Training on conscious technology use should be provided	16	3
Organizing the environment	Engaging activities should be organized at schools	27	9
	Communication in families should be prioritized	12	9
	Family elders (parents) should set an example	8	6
	Educational platforms should be created	5	3
	Student-centered activities should be implemented in lessons	5	1
	Teachers should be guides and role models	4	2
	Students should be given opportunities to work together	4	1
	Less emphasis should be placed on digital activities in lessons	2	1
	It should be banned at school	1	-
Usage control	Time limitations should be set	31	6
	Introducing screens later in childhood	7	1
	Digital detox	7	1
	Schools and families should collaborate	6	2
Developing individual alternatives	Develop hobbies (Art, sports, culture)	59	12
	Develop a sense of responsibility	8	4
	Gain self-regulation skills	5	2
	Change lifestyle when necessary	2	1

digital use. Many participants underscored the need for engaging, creative school activities that support students' personal development. One participant noted:

Schools should design activities that boost students' self-confidence and catch their interest. Activities should encourage creativity. (D22).

Additionally, strong family communication and parental role modelling were frequently cited as preventive factors. As one participant reflected:

Families should put down their phones and spend quality time with their children. They can do art activities together or read books. (K22).

Several students also recommended the development of educational platforms aimed at fostering healthier digital habits (D6). Relatedly, student-centered classroom practices that promote collaboration and active learning

were also considered beneficial. For instance, participants suggested implementing debates, problem-solving tasks, and experimental learning to enhance lesson engagement (D54, D76). Teachers were also seen as pivotal role models in this process as stated by a participant:

Teachers and parents should be role models for students. (D15).

In addition, participants raised concerns about the increasing reliance on digital tools within the classroom. Some students recommended reducing digital activities in lessons to minimize distractions and addictive behaviors. One participant explained her experience:

Our professors give assignments, I have to check my phone, research on the internet, sometimes I even read the book on the screen. But then I end up spending hours hovering [around] useless websites. (D12)

Under the theme of usage control, participants called for setting screen time limits, especially enforced by families (K22). Others recommended introducing children to digital devices at a later age and limiting their use once introduced (D12). The idea of "digital detox" also emerged, with participants encouraging outdoor activities and time away from screens:

Digital detox is essential, we should go outside, take walks, and pick up different hobbies. (D34)

In this context, school-family cooperation was considered vital. As one participant noted:

Both the school and the family have a lot of work to do to prevent brain rot. Only if they work together can this problem be solved. (D82)

Finally, in terms of developing individual alternatives, students emphasized cultivating hobbies (e.g., art, sports, culture) and participating in social activities. These were seen not only as diversions from digital overuse but also as meaningful ways to enhance well-being. One participant shared his suggestions:

New hobbies can be found and done, trips can be organized, socializing can be increased, art can be practiced. For example, I play the guitar and sing which helps me prevent brain rot. (K10)

Other students stressed the importance of building responsibility and self-regulation (D23, K14), while a few advocated for broader lifestyle changes. One participant highlighted:

To get rid of brain rot, we should live an active life, take care of our bodies and souls, pick up different activities, and live life to the fullest. (D3)

Discussion

The study suggests that preservice teachers in the faculty of education may exhibit patterns of cognitive fatigue and reduced attention span, potentially influenced by a combination of emotional, habitual, environmental, social, and cognitive factors. Emotional drivers such as the desire to relieve boredom, the tendency to avoid responsibilities, the sense of incompleteness when disconnected from social media, the need to escape real-life challenges, fear of missing out (FOMO), and the pursuit of happiness through online platforms emerged as key contributors. Uğurlu [44] similarly found that boredom during free time significantly predicts social media addiction among university students. Twenge and Campbell [45] argued that digital addiction functions as an emotional coping mechanism among youth, echoing the present study's findings. Przybylski et al. [46] introduced the concept of FOMO to describe the persistent concern about missing out on social interactions, which aligns with the participants' reported anxiety when offline. In addition, Meng and Leung [8] found that escapism and entertainment motives drive excessive use of platforms like TikTok, reinforcing the emotional underpinnings of brain rot observed in this study.

With regard to habitual factors, students reported using digital devices to fill free time, entertain themselves, relax, and pursue convenience often in repetitive and unconscious ways. Notifications and likes in digital content cause the release of dopamine. Dopamine brings about the expectation of reward and, consequently, behavioral addiction [47]. These findings reflect Duhigg's [48] habit loop theory, which explains how behaviors become automatic through cue-routine-reward cycles. The recurring use of digital media in idle moments suggests that brain rot, for many students, has become a deeply ingrained habit. This pattern aligns with [5], who found that entertainment motives are significant predictors of social media dependency. Uğurlu [44] further confirmed that increased free time is associated with greater risk of social media addiction.

Environmental and social factors also contributed substantially to brain rot. Participants cited the COVID-19 pandemic, peer influence, the digitalization of education, and the accessibility and appeal of digital content as major drivers. UNESCO [49] similarly reported that the pandemic accelerated digital device dependency among youth, increasing vulnerability to addictive behaviors. Büyükgebiz Koca and Tunca [14] found that loneliness during the pandemic was linked to pathological internet use among young people, corroborating this study's

results. Students' efforts to maintain online visibility and conform to digital norms reflect Festinger's [50] Social Comparison Theory, which posits those individuals evaluate themselves in comparison to others—often using social media as the primary reference point. Likewise, Balci et al. [5] emphasized that platforms like TikTok are intentionally designed to maximize engagement and dependency. Cognitive contributors such as mental fatigue, heightened curiosity, and the desire for immediate access to information also fueled brain rot. Dinç [51] identified curiosity as a central driver of internet addiction, while Kızılkaya [11] found that the search for psychological relief contributes to problematic digital use both findings consistent with this study.

Brain rot was also found to significantly affect students' educational experiences, particularly in terms of educational readiness and productivity. This study provides pre-service teachers with a valuable opportunity to enhance their awareness of brain rot and its potential impact on the educational process, thereby equipping them with an advantage in addressing related challenges. This research allows pre-service teachers to gain an important advantage in overcoming problems caused by brain rot in the educational process by increasing their level of awareness of this concept [52]. Such awareness is crucial, as teachers who understand the cognitive and emotional effects of brain rot can act as positive role models for their future students, promoting healthier and more balanced digital behaviors. The findings, particularly the frequent reports of attention deficits, mental fog, and emotional difficulties among participants, underscore the importance of integrating digital well-being education into teacher training programmes. Kandasamy et al. [17] identified similar effects in their work on internet addiction, highlighting mental, emotional, and physical consequences. Within the scope of educational readiness, brain rot causes cognitive decline, attention deficit, mental fatigue, and foggy. According to social cognitive theory, paying attention and encoding into memory are very important in the process of developing behavior [53]. Unlike the results of this study, Lodge and Harrison [54] mention in their research that the effect of technology on attention is not unidirectional; for example, some video games may improve visual attention skills. The researchers revealed, in the context of attention learning theory, that attention is not a fixed ability and can be shaped through experience and the environment. Ophir et al. [55] found that constant exposure to digital media diminishes cognitive control and concentration. Recent findings by Lakilaki et al. [56] reinforce these conclusions, showing that digital overexposure impairs attention and focus in academic settings. Consistent with the findings of this study, Sparrow et al. [57] demonstrated that the widespread use of the internet

fosters a form of cognitive offloading, leading to cognitive laziness and a diminished capacity for deep information processing. Similarly, Carr [52] argued that extensive internet use encourages superficial learning patterns and contributes to forgetfulness by weakening long-term memory consolidation. Recent research on brain rot further supports these perspectives, indicating that the phenomenon emerges as a consequence of excessive digital content consumption, which disrupts sustained attention and cognitive engagement. In this context, Adam & Cik Soh [31] revealed in their research that brain rot caused by excessive screen time can lead to problems such as memory issues, decreased attention span, and impaired decision-making. Nayak [32] noted that the concept of brain rot has become increasingly widespread in recent years and defined brain rot, as also demonstrated in this study, as a decline in cognitive functions and attention span. This decline caused by brain rot can lead to more serious problems, such as the transformation of cultural richness into consumable commodities and the superficialization of cultural wealth [52].

The emotional toll of brain rot in this study marked by symptoms of depression, anxiety, emptiness, guilt, addiction, burnout, and reduced self-confidence echoes findings by Gökbulut and Bilen [58], who demonstrated that social media addiction correlates with elevated stress, anxiety, and depressive symptoms. Shen et al. [13] and Jaishy et al. [59] similarly reported that dysfunctional internet use leads to significant psychological distress. In their study on brain rot, Yousef et al. [33] concluded that brain rot has negative effects on mental health. These results indicate that brain rot may have long-term effects.

Physically, students described symptoms such as insomnia, neck and back pain, and eye strain conditions supported by Cain and Gradisar [60], who found that increased screen time disrupts sleep cycles, and by Achangwa et al. [61], who reported musculoskeletal discomfort among those with smartphone addiction. Yazgan [62] conducted a study on brain rot and found that the condition leads to sleeplessness and adversely affects learning and memory processes.

From an educational productivity perspective, brain rot was found to reduce students' ability to comprehend course material, retain information, and engage in deep learning, while also contributing to procrastination and academic underperformance. Rathakrishnan et al. [16] similarly concluded that smartphone addiction negatively impacts academic achievement. Kirschner and Karpinski [22] found that frequent use of platforms like Facebook correlates with reduced study time and poorer academic outcomes. Sun and Chao [63] further demonstrated that multitasking with digital media leads to fragmented attention and diminished academic performance. These results align with Aznar-Díaz et al. [19], who identified

a strong link between problematic internet use and academic procrastination.

Socially, participants reported difficulties in maintaining peer relationships and initiating face-to-face interactions, contributing to isolation and reluctance to attend school. Han, Nagduar, and Yu [21] showed that digital addiction impairs interpersonal communication, while Achangwa et al. [61] highlighted its role in reducing real-life social engagement. Students in this study expressed reduced motivation to attend classes or engage in school activities, echoing Akbaba and Kaya's [15] finding that problematic internet use lowers academic motivation. Finally, brain rot was found to disrupt students' sense of time and time management. Gök [64] reported that students often use digital tools inefficiently, leading to significant time loss. Jojo and Sundaramoorthy [65] emphasized that weak time management skills often tied to a lack of personal responsibility are associated with excessive internet use, supporting this study's findings. Owens [66] reported findings that diverge from those of the present study, concluding that young people perceive brain rot as digital engagement, and that the time spent in this process cannot be considered leisure time, as they believe this time provides them with relaxation. In addition, Owens associated brain rot with earlier forms of purposeless entertainment, such as "killing time" or passively watching television, arguing that the phenomenon should not be regarded as entirely new.

In response to these challenges, participants proposed several strategies to prevent brain rot, including educational interventions, environmental restructuring, usage control, and the development of personal alternatives. Education was seen as a crucial tool in combating brain rot, with calls for digital literacy training, awareness campaigns on digital addiction, and guidance on responsible technology use. Aribowo and Bagaskara [67] found in a way that supports the results of this study, that brain rot is not only an individual problem but a structural issue produced by the digital ecology, and therefore, measures need to be taken at the societal level. These suggestions align with Achangwa et al. [61], who advocated for targeted education to mitigate smartphone addiction.

Participants also emphasized the importance of family dynamics and parental modeling [68] highlighted that internet addiction strains family relationships and that improved communication can serve as a protective factor. Recommendations for limiting screen time and implementing digital detox practices were also raised. Young [27] demonstrated the effectiveness of time restrictions in curbing internet addiction, while Varma [69] recommended structured detox periods as a preventive strategy.

In terms of educational practices, participants advised reducing digital content in classrooms, arguing that it

exacerbates brain rot. Miniakhmetova et al. [70] supported this concern, noting that increased digital learning may heighten students' susceptibility to digital dependency. Student-centered, collaborative classroom strategies were proposed as alternatives. Bağcı [1] similarly recommended engaging teaching methods to reduce digital reliance among university students.

Finally, students proposed developing personal alternatives to screen use, such as engaging in arts, sports, and cultural activities, as well as participating in social events. These strategies were complemented by calls for increased self-regulation and personal responsibility. Wibowo et al. [59] found that individuals with higher self-regulation are less susceptible to digital addiction, while Lakilaki et al. [56] observed that outdoor and physical activities can mitigate the effects of brain rot. Together, these findings underscore the importance of both systemic interventions and personal coping strategies in addressing the cognitive and educational risks associated with brain rot.

Recommendations

Schools could implement phone-use restrictions during specific hours or lessons to minimize digital distraction and support sustained attention. The use of applications that monitor students' daily screen time should be encouraged and made more widespread. Educators can foster deep reading habits by encouraging students to engage with printed materials more frequently and to practice slow, reflective reading to strengthen cognitive focus and comprehension.

This study lays the groundwork for quantitative research aimed at determining the wider prevalence of the "brain rot" phenomenon or for longitudinal studies that aim to track the long-term effects of digital habits. Furthermore, future cross-cultural comparative studies would provide valuable insights by revealing how digital behaviors and the manifestations of brain rot vary across different cultural and educational contexts.

Limitations

The study was conducted only at a single university in Türkiye; therefore, the findings may not be generalizable to broader populations or different cultural contexts. Although data triangulation was employed, the screen-time data derived from participants' self-reports may be subject to bias, as participants might not have reported their digital habits with complete accuracy. Furthermore, as the researcher actively conducted observations throughout the study, this involvement may have introduced subjective influences during the interpretation process, making it challenging to entirely eliminate researcher bias.

Although the study was conducted with 113 participants, in-depth interviews were carried out with only 30 individuals. While the findings indicate that the majority of participants exhibited symptoms of "brain rot," detailed qualitative insights into the experiences of those who were not interviewed remained limited. This limitation was partially mitigated through the use of data triangulation.

Conclusions

This study on brain rot, a phenomenon emerging among students as a consequence of digital addiction, revealed that the condition arises from a complex interplay of emotional, habitual, environmental, social, and cognitive factors. Emotional factors included boredom and the desire for relaxation, avoidance of tasks or inactivity, feelings of deficiency when not engaging with social media, escapism from real-life problems, and happiness derived from online interactions. Habitual tendencies such as filling free time, seeking entertainment or rest, and opting for convenience were also influential. Additionally, environmental, social, and cognitive reasons—such as the pandemic period, peer pressure, the digitalization of education, the appeal and accessibility of digital content, the desire for individuality, and anxieties regarding communication and visibility further contributed to the development of brain rot. The findings indicate that brain rot manifests in educational contexts in two primary dimensions: educational readiness and educational efficiency. In terms of readiness, students suffering from brain rot reported attention deficits, mental fog, depression, anxiety, feelings of emptiness, regret or guilt, burnout, loss of self-confidence, impatience, social withdrawal, loneliness, difficulty focusing on lessons, cognitive decline, mental fatigue, forgetfulness, sleep disturbances, and physical symptoms such as eye strain and neck or back pain. In relation to educational efficiency, the condition led to challenges in comprehension and learning, a decline in academic achievement, procrastination, weakened social relationships, reluctance to attend school, and reduced engagement in educational activities. Furthermore, brain rot was found to undermine students' time management, resulting in significant time loss and a distorted perception of time.

Abbreviations

FOMO	Fear of Missing Out
UNESCO	United Nations Educational, Scientific and Cultural Organization

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Consent to participate declaration

All participants were informed about the purpose, procedure, and their rights regarding the study. They voluntarily agreed to participate and provided informed consent prior to data collection.

Authors' contributions

The sole author was responsible for all aspects of this study, including conceptualization, data collection, analysis, and manuscript preparation.

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Data availability

Data are kept by the authors and can be shared upon request.

Declarations

Ethics approval and consent to participate

The approval request for this research has been submitted to the Rectorate of Kahramanmaraş Sütçü İmam University – Social and Human Sciences Ethics Committee on 12 February 2025. This research has been approved by the aforementioned committee on 27 March 2025, with reference number E-72321963-050.04-405671. The study was conducted in accordance with Decision No. 21, taken at the meeting dated March 21, 2025 (Session No. 2025-6) of the Ethics Committee for Social and Human Sciences at Kahramanmaraş Sütçü İmam University, and within the framework of the APA ethical principles. All procedures performed in this study were in accordance with the ethical standards of the institutional committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Consent for publication

The structure of this study makes consent for publication not applicable.

Competing interests

The authors declare no competing interests.

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